

[7/28] 스마트 데이터 이해 및 활용 _Day2_실습

[GROUP BY 1]

	AZ 부서명(department_name)	123 부서별 직원 수
1	Sales	11
2	Finance	6
3	Purchasing	6
4	IT	5
5	Executive	3
6	R&D Korea	3
7	Shipping	3
8	Accounting	2
9	Marketing	2
10	부서 미배정	1
11	Administration	1
12	Human Resources	1
13	Public Relations	1

[GROUP BY 2]

	AZ 직무명(job_title)	123 직무별 직원 수	123 직무별 평균 급여
1	President	1	24,000
2	Column: 직무명(job_title) varchar(35) NOT NULL	2	17,000
3	Table: skala_db.hr.jobs	1	13,000
4	Read-only: No unique key. Row modification is not available.	5	12,200
5	Sales Manager	1	12,000
6	Accounting Manager	1	12,000
7	Finance Manager	1	11,000
8	Purchasing Manager	1	10,000
9	Public Relations Representative	6	8,500
10	Sales Representative	1	8,300
11	Public Accountant	3	8,033.33
12	Stock Manager	5	7,920
13	Accountant	8	6,950
14	Programmer	1	6,500
15	Human Resources Representative	2	6,100
16	Marketing Representative	1	4,400
17	Administration Assistant	5	2,780
18	Purchasing Clerk		

[GROUP BY 3]

	AZ 부서명(department_name)	123 부서별 급여 합계	123 부서별 급여 평균	123 부서별 급여 최솟값	123 부서별 급여 최댓값
1	Sales	112,000	10,181.82	7,000	14,000
2	Executive	58,000	19,333.33	17,000	24,000
3	Finance	51,600	8,600	6,900	12,000
4	IT	28,800	5,760	4,200	9,000
5	R&D Korea	26,800	8,933.33	7,800	10,500
6	Purchasing	24,900	4,150	2,500	11,000
7	Shipping	24,100	8,033.33	7,900	8,200
8	Accounting	20,300	10,150	8,300	12,000
9	Marketing	19,000	9,500	6,000	13,000
10	Public Relations	10,000	10,000	10,000	10,000
11	Human Resources	6,500	6,500	6,500	6,500
12	부서 미배정	6,200	6,200	6,200	6,200
13	Administration	4,400	4,400	4,400	4,400

[GROUP BY 4]

	123 관리자 사번(employee_id)	AZ 관리자명(employee_name)	123 관리자별 부하 직원 수
1	100	Steven King	12
2	145	John Russell	6
3	101	Neena Kochhar	5
4	108	Nancy Greenberg	5
5	114	Den Raphaely	5
6	103	Alexander Hunold	4
7	102	Lex De Haan	2
8	201	Michael Hartstein	2
9	220	Minjun Kim	2
10	205	Shelley Higgins	1

[GROUP BY 5]

	123 입사 연도(hire_year)	123 입사 연도별 입사자 수
1	2,001	1
2	2,002	7
3	2,003	4
4	2,004	3
5	2,005	13
6	2,006	6
7	2,007	6
8	2,008	1
9	2,019	1
10	2,020	1
11	2,021	1
12	2,023	1

[GROUP BY 6]

	AZ 커미션 수령 여부 ▼	123 커미션 수령 여부별 인원 ▼	123 커미션 수령 여부별 평균 급여 ▼
1	수령	11	10,181.82
2	미수령	34	8,252.94

[GROUP BY 7]

	AZ 국가명(country_name) ▼	123 국가별 근무 인원 수 ▼
1	United States	26
2	United Kingdom	12
3	South Korea	3
4	Canada	2
5	Germany	1

[GROUP BY 8]

	AZ 부서명(department_name) ▼	123 부서별 직무 종류 수 ▼
1	Accounting	2
2	Executive	2
3	Finance	2
4	Marketing	2
5	Purchasing	2
6	Sales	2
7	부서 미배정	1
8	Administration	1
9	Human Resources	1
10	IT	1
11	Public Relations	1
12	R&D Korea	1
13	Shipping	1

[GROUP BY 9]

	AZ 부서명(department_name) ▼	123 부서별 평균 급여 ▼
1	Executive	19,333.33
2	Sales	10,181.82
3	Accounting	10,150
4	Public Relations	10,000
5	Marketing	9,500
6	R&D Korea	8,933.33
7	Finance	8,600
8	Shipping	8,033.33

[GROUP BY 10]

	AZ 부서명 (department_name) ▼	AZ 급여 구간 ▼	123 부서별 급여 구간 인원수 ▼
1	부서 미배정	중위 (5000-9999)	1
2	Accounting	상위 (10000+)	1
3	Accounting	중위 (5000-9999)	1
4	Administration	하위 (5000미만)	1
5	Executive	상위 (10000+)	3
6	Finance	상위 (10000+)	1
7	Finance	중위 (5000-9999)	5
8	Human Resources	중위 (5000-9999)	1
9	IT	중위 (5000-9999)	2
10	IT	하위 (5000미만)	3
11	Marketing	상위 (10000+)	1
12	Marketing	중위 (5000-9999)	1
13	Public Relations	상위 (10000+)	1
14	Purchasing	상위 (10000+)	1
15	Purchasing	하위 (5000미만)	5
16	R&D Korea	상위 (10000+)	1
17	R&D Korea	중위 (5000-9999)	2
18	Sales	상위 (10000+)	6
19	Sales	중위 (5000-9999)	5
20	Shipping	중위 (5000-9999)	3

[서브쿼리 1]

	123 사번(employee_id)	AZ 사원명(employee_name)	123 급여(salary)
1	100	Steven King	24,000
2	102	Lex De Haan	17,000
3	101	Neena Kochhar	17,000
4	145	John Russell	14,000
5	146	Karen Partners	13,500
6	201	Michael Hartstein	13,000
7	147	Alberto Errazuriz	12,000
8	205	Shelley Higgins	12,000
9	108	Nancy Greenberg	12,000
10	148	Gerald Cambrault	11,000
11	114	Den Raphaely	11,000
12	149	Eleni Zlotkey	10,500
13	220	Minjun Kim	10,500
14	150	Peter Tucker	10,000
15	204	Hermann Baer	10,000
16	151	David Bernstein	9,500
17	152	Peter Hall	9,000
18	103	Alexander Hunold	9,000
19	109	Daniel Faviat	9,000

[서브쿼리 2]

	123 사번(employee_id)	AZ 사원명(employee_name)	123 부서 번호(department_id)	123 급여(salary)
1	201	Michael Hartstein	20	13,000
2	114	Den Raphaely	30	11,000
3	121	Adam Fripp	50	8,200
4	103	Alexander Hunold	60	9,000
5	104	Bruce Ernst	60	6,000
6	145	John Russell	80	14,000
7	146	Karen Partners	80	13,500
8	147	Alberto Errazuriz	80	12,000
9	148	Gerald Cambrault	80	11,000
10	149	Eleni Zlotkey	80	10,500
11	100	Steven King	90	24,000
12	108	Nancy Greenberg	100	12,000
13	109	Daniel Faviat	100	9,000
14	205	Shelley Higgins	110	12,000
15	220	Minjun Kim	120	10,500

[서브쿼리 3]

	123 부서 번호(department_id)	A-Z 부서명(department_name)
1	10	Administration
2	20	Marketing
3	30	Purchasing
4	40	Human Resources
5	50	Shipping
6	60	IT
7	70	Public Relations
8	80	Sales
9	90	Executive
10	100	Finance
11	110	Accounting
12	120	R&D Korea

[서브쿼리 4]

	123 부서 번호(department_id)	A-Z 부서명(department_name)	123 부서별 평균 급여
1	90	Executive	19,333.33

[서브쿼리 5]

	123 사번(employee_id)	AZ 사원명(employee_name)
1	100	Steven King
2	103	Alexander Hunold
3	104	Bruce Ernst
4	105	David Austin
5	106	Valli Pataballa
6	107	Diana Lorentz
7	108	Nancy Greenberg
8	109	Daniel Faviet
9	110	John Chen
10	111	Ismael Sciarra
11	112	Jose Manuel Urman
12	113	Luis Popp
13	115	Alexander Khoo
14	116	Shelli Baida
15	117	Sigal Tobias
16	118	Guy Himuro
17	119	Karen Colmenares
18	120	Matthew Weiss
19	121	Adam Fripp
20	145	John Russell
21	146	Karen Partners
22	147	Alberto Errazuriz
23	148	Gerald Cambrault
24	149	Eleni Zlotkey
25	150	Peter Tucker
26	151	David Bernstein
27	152	Peter Hall
28	153	Christopher Olsen
29	154	Nanette Cambrault
30	155	Oliver Tuvault
31	202	Pat Fay
32	203	Susan Mavris
33	204	Hermann Baer
34	205	Shelley Higgins
35	206	William Gietz
36	221	Seoyeon Lee
37	222	Jihoon Park

[CTE 1]

	123 부서 번호(department_id)	AZ 부서명(department_name)	123 부서별 평균 급여	123 전체 평균 급여	AZ 비교 결과
1	90	Executive	19,333.33	8,724.44	전체 평균 이상
2	80	Sales	10,181.82	8,724.44	전체 평균 이상
3	110	Accounting	10,150	8,724.44	전체 평균 이상
4	70	Public Relations	10,000	8,724.44	전체 평균 이상
5	20	Marketing	9,500	8,724.44	전체 평균 이상
6	120	R&D Korea	8,933.33	8,724.44	전체 평균 이상
7	100	Finance	8,600	8,724.44	전체 평균 미만
8	50	Shipping	8,033.33	8,724.44	전체 평균 미만
9	40	Human Resources	6,500	8,724.44	전체 평균 미만
10	60	IT	5,760	8,724.44	전체 평균 미만
11	10	Administration	4,400	8,724.44	전체 평균 미만
12	30	Purchasing	4,150	8,724.44	전체 평균 미만

[CTE 2]

	123 부서 번호(department_id)	AZ 부서명(department_name)	123 사번(employee_id)	AZ 사원명(employee_name)	123 급여(salary)
1	10	Administration	200	Jennifer Whalen	4,400
2	20	Marketing	201	Michael Hartstein	13,000
3	30	Purchasing	114	Den Raphaely	11,000
4	40	Human Resources	203	Susan Mavris	6,500
5	50	Shipping	121	Adam Fripp	8,200
6	60	IT	103	Alexander Hunold	9,000
7	70	Public Relations	204	Hermann Baer	10,000
8	80	Sales	145	John Russell	14,000
9	90	Executive	100	Steven King	24,000
10	100	Finance	108	Nancy Greenberg	12,000
11	110	Accounting	205	Shelley Higgins	12,000
12	120	R&D Korea	220	Minjun Kim	10,500

[CTE 3] - 데이터 45개

	123 조직 단계(level)	123 사번(employee_id)	AZ 사원명(employee_name)	123 관리자 사번(manager_id)
1	1	100	Steven King	[NULL]
2	2	101	Neena Kochhar	100
3	2	102	Lex De Haan	100
4	2	114	Den Raphaely	100
5	2	120	Matthew Weiss	100
6	2	121	Adam Fripp	100
7	2	122	Payam Kaufling	100
8	2	145	John Russell	100
9	2	146	Karen Partners	100
10	2	147	Alberto Errazuriz	100
11	2	148	Gerald Cambrault	100
12	2	149	Eleni Zlotkey	100
13	2	201	Michael Hartstein	100
14	3	103	Alexander Hunold	102
15	3	108	Nancy Greenberg	101
16	3	115	Alexander Khoo	114
17	3	116	Shelli Baida	114
18	3	117	Sigal Tobias	114
19	3	118	Guy Himuro	114
20	3	119	Karen Colmenares	114
21	3	150	Peter Tucker	145
22	3	151	David Bernstein	145
23	3	152	Peter Hall	145
24	3	153	Christopher Olsen	145
25	3	154	Nanette Cambrault	145
26	3	155	Oliver Tuvault	145
27	3	200	Jennifer Whalen	101
28	3	202	Pat Fay	201
29	3	203	Susan Mavris	101
30	3	204	Hermann Baer	101
31	3	205	Shelley Higgins	101
32	3	220	Minjun Kim	102
33	3	223	Yuna Choi	201
34	4	104	Bruce Ernst	103
35	4	105	David Austin	103
36	4	106	Valli Pataballa	103
37	4	107	Diana Lorentz	103
38	4	109	Daniel Faviat	108
39	4	110	John Chen	108
40	4	111	Ismael Sciarra	108
41	4	112	Jose Manuel Urman	108
42	4	113	Luis Popp	108
43	4	206	William Gietz	205

[JOIN 1] 데이터 44개

	123 → 사번(employee_id)	AZ 사원명(employee_name)	123 → 부서 번호(department_id)	AZ 부서명(department_name)
1	100	Steven King	90	Executive
2	101	Neena Kochhar	90	Executive
3	102	Lex De Haan	90	Executive
4	103	Alexander Hunold	60	IT
5	104	Bruce Ernst	60	IT
6	105	David Austin	60	IT
7	106	Valli Pataballa	60	IT
8	107	Diana Lorentz	60	IT
9	108	Nancy Greenberg	100	Finance
10	109	Daniel Faviet	100	Finance
11	110	John Chen	100	Finance
12	111	Ismael Sciarra	100	Finance
13	112	Jose Manuel Urman	100	Finance
14	113	Luis Popp	100	Finance
15	114	Den Raphaely	30	Purchasing
16	115	Alexander Khoo	30	Purchasing
17	116	Shelli Baida	30	Purchasing
18	117	Sigal Tobias	30	Purchasing
19	118	Guy Himuro	30	Purchasing
20	119	Karen Colmenares	30	Purchasing
21	120	Matthew Weiss	50	Shipping

[JOIN 2] 데이터 45개

	123 → 사번(employee_id)	AZ 사원명(employee_name)	AZ → 직무 코드(job_id)	AZ 직무명(job_title)
1	100	Steven King	AD_PRES	President
2	101	Neena Kochhar	AD_VP	Administration Vice President
3	102	Lex De Haan	AD_VP	Administration Vice President
4	103	Alexander Hunold	IT_PROG	Programmer
5	104	Bruce Ernst	IT_PROG	Programmer
6	105	David Austin	IT_PROG	Programmer
7	106	Valli Pataballa	IT_PROG	Programmer
8	107	Diana Lorentz	IT_PROG	Programmer
9	108	Nancy Greenberg	FI_MGR	Finance Manager
10	109	Daniel Faviet	FI_ACCOUNT	Accountant
11	110	John Chen	FI_ACCOUNT	Accountant
12	111	Ismael Sciarra	FI_ACCOUNT	Accountant
13	112	Jose Manuel Urman	FI_ACCOUNT	Accountant
14	113	Luis Popp	FI_ACCOUNT	Accountant
15	114	Den Raphaely	PU_MAN	Purchasing Manager
16	115	Alexander Khoo	PU_CLERK	Purchasing Clerk
17	116	Shelli Baida	PU_CLERK	Purchasing Clerk
18	117	Sigal Tobias	PU_CLERK	Purchasing Clerk
19	118	Guy Himuro	PU_CLERK	Purchasing Clerk
20	119	Karen Colmenares	PU_CLERK	Purchasing Clerk
21	120	Matthew Weiss	ST_MAN	Stock Manager

[JOIN 3] 데이터 44개

	123 사번(employee_id)	AZ 사원명(employee_name)	AZ 부서명(department_name)	AZ 도시(city)	AZ 국가명(country_name)
1	201	Michael Hartstein	Marketing	Toronto	Canada
2	202	Pat Fay	Marketing	Toronto	Canada
3	204	Hermann Baer	Public Relations	Munich	Germany
4	220	Minjun Kim	R&D Korea	Seoul	South Korea
5	221	Seoyeon Lee	R&D Korea	Seoul	South Korea
6	222	Jihoon Park	R&D Korea	Seoul	South Korea
7	145	John Russell	Sales	Oxford	United Kingdom
8	146	Karen Partners	Sales	Oxford	United Kingdom
9	147	Alberto Errazuriz	Sales	Oxford	United Kingdom
10	148	Gerald Cambraut	Sales	Oxford	United Kingdom
11	149	Eleni Zlotkey	Sales	Oxford	United Kingdom
12	150	Peter Tucker	Sales	Oxford	United Kingdom
13	151	David Bernstein	Sales	Oxford	United Kingdom
14	152	Peter Hall	Sales	Oxford	United Kingdom
15	153	Christopher Olsen	Sales	Oxford	United Kingdom
16	154	Nanette Cambraut	Sales	Oxford	United Kingdom
17	155	Oliver Tuvault	Sales	Oxford	United Kingdom
18	203	Susan Mavris	Human Resources	London	United Kingdom
19	100	Steven King	Executive	Seattle	United States
20	101	Neena Kochhar	Executive	Seattle	United States
21	102	Lex De Haan	Executive	Seattle	United States

[JOIN 4] 데이터 44개

	123 사번(employee_id)	AZ 사원명(employee_name)	123 관리자 사번(manager_id)	AZ 관리자명(manager_name)
1	101	Neena Kochhar	100	Steven King
2	102	Lex De Haan	100	Steven King
3	103	Alexander Hunold	102	Lex De Haan
4	104	Bruce Ernst	103	Alexander Hunold
5	105	David Austin	103	Alexander Hunold
6	106	Valli Pataballa	103	Alexander Hunold
7	107	Diana Lorentz	103	Alexander Hunold
8	108	Nancy Greenberg	101	Neena Kochhar
9	109	Daniel Faviat	108	Nancy Greenberg
10	110	John Chen	108	Nancy Greenberg
11	111	Ismael Sciarra	108	Nancy Greenberg
12	112	Jose Manuel Urman	108	Nancy Greenberg
13	113	Luis Popp	108	Nancy Greenberg
14	114	Den Raphaely	100	Steven King
15	115	Alexander Khoo	114	Den Raphaely
16	116	Shelli Baida	114	Den Raphaely
17	117	Sigal Tobias	114	Den Raphaely
18	118	Guy Himuro	114	Den Raphaely
19	119	Karen Colmenares	114	Den Raphaely
20	120	Matthew Weiss	100	Steven King
21	121	Adam Friess	100	Steven King

[JOIN 5] 데이터 45개

	123 사번(employee_id)	A-Z 사원명(employee_name)	123 부서 번호(department_id)	A-Z 부서명(department_name)
1	100	Steven King	90	Executive
2	101	Neena Kochhar	90	Executive
3	102	Lex De Haan	90	Executive
4	103	Alexander Hunold	60	IT
5	104	Bruce Ernst	60	IT
6	105	David Austin	60	IT
7	106	Valli Pataballa	60	IT
8	107	Diana Lorentz	60	IT
9	108	Nancy Greenberg	100	Finance
10	109	Daniel Faviet	100	Finance
11	110	John Chen	100	Finance
12	111	Ismael Sciarra	100	Finance
13	112	Jose Manuel Urman	100	Finance
14	113	Luis Popp	100	Finance
15	114	Den Raphaely	30	Purchasing
16	115	Alexander Khoo	30	Purchasing
17	116	Shelli Baida	30	Purchasing
18	117	Sigal Tobias	30	Purchasing
19	118	Guy Himuro	30	Purchasing
20	119	Karen Colmenares	30	Purchasing
21	120	Matthew Weiss	50	Shipping

[JOIN 6] 데이터 44개

	123 부서 번호(department_id)	A-Z 부서명(department_name)	123 사번(employee_id)	A-Z 사원명(employee_name)
1	10	Administration	200	Jennifer Whalen
2	20	Marketing	201	Michael Hartstein
3	20	Marketing	202	Pat Fay
4	30	Purchasing	114	Den Raphaely
5	30	Purchasing	115	Alexander Khoo
6	30	Purchasing	116	Shelli Baida
7	30	Purchasing	117	Sigal Tobias
8	30	Purchasing	118	Guy Himuro
9	30	Purchasing	119	Karen Colmenares
10	40	Human Resources	203	Susan Mavris
11	50	Shipping	120	Matthew Weiss
12	50	Shipping	121	Adam Fripp
13	50	Shipping	122	Payam Kaufling
14	60	IT	103	Alexander Hunold
15	60	IT	104	Bruce Ernst
16	60	IT	105	David Austin
17	60	IT	106	Valli Pataballa
18	60	IT	107	Diana Lorentz
19	70	Public Relations	204	Hermann Baer
20	80	Sales	145	John Russell
21	90	Sales	146	Karen Partners

[JOIN 7] 데이터 47개

	123 사번 (employee_id)	AZ 사원명 (employee_name)	AZ 직무 코드 (job_id)	AZ 직무명 (job_title)	
1	206	William Gietz	AC_ACCOUNT	Public Accountant	
2	205	Shelley Higgins	AC_MGR	Accounting Manager	
3	200	Jennifer Whalen	AD_ASST	Administration Assistant	
4	100	Steven King	AD PRES	President	
5	101	Neena Kochhar	AD_VP	Administration Vice President	
6	102	Lex De Haan	AD_VP	Administration Vice President	
7	109	Daniel Faviet	FL_ACCOUNT	Accountant	
8	110	John Chen	FL_ACCOUNT	Accountant	
9	111	Ismael Sciarra	FL_ACCOUNT	Accountant	
10	112	Jose Manuel Urman	FL_ACCOUNT	Accountant	
11	113	Luis Popp	FL_ACCOUNT	Accountant	
12	108	Nancy Greenberg	FL_MGR	Finance Manager	
13	203	Susan Mavris	HR_REP	Human Resources Representative	
14	103	Alexander Hunold	IT_PROG	Programmer	
15	104	Bruce Ernst	IT_PROG	Programmer	
16	105	David Austin	IT_PROG	Programmer	
17	106	Valli Pataballa	IT_PROG	Programmer	
18	107	Diana Lorentz	IT_PROG	Programmer	
19	220	Minjun Kim	IT_PROG	Programmer	
20	221	Seoyeon Lee	IT_PROG	Programmer	
21	222	Ilhoon Park	IT_PROG	Programmer	

[JOIN 8] 데이터 45개

	123 사번 (employee_id)	AZ 사원명 (employee_name)	123 부서 번호 (department_id)	AZ 부서명 (department_name)	
1	200	Jennifer Whalen	10	Administration	
2	201	Michael Hartstein	20	Marketing	
3	202	Pat Fay	20	Marketing	
4	114	Den Raphaely	30	Purchasing	
5	115	Alexander Khoo	30	Purchasing	
6	116	Shelli Baida	30	Purchasing	
7	117	Sigal Tobias	30	Purchasing	
8	118	Guy Himuro	30	Purchasing	
9	119	Karen Colmenares	30	Purchasing	
10	203	Susan Mavris	40	Human Resources	
11	120	Matthew Weiss	50	Shipping	
12	121	Adam Fripp	50	Shipping	
13	122	Payam Kaufling	50	Shipping	
14	103	Alexander Hunold	60	IT	
15	104	Bruce Ernst	60	IT	
16	105	David Austin	60	IT	
17	106	Valli Pataballa	60	IT	
18	107	Diana Lorentz	60	IT	
19	204	Hermann Baer	70	Public Relations	
20	145	John Russell	80	Sales	
21	146	Karen Partners	80	Sales	

[JOIN 9]

	123 사번 (employee_id)	AZ 사원명 (employee_name)	시작일 (start_date)	종료일 (end_date)	AZ 직무명 (job_title)	AZ 부서명 (department_name)
1	101	Neena Kochhar	1997-09-21 GMT+09:00	2001-10-27	Public Accountant	Accounting
2	101	Neena Kochhar	2001-10-28 GMT+09:00	2005-03-15	Accounting Manager	Accounting
3	102	Lex De Haan	1993-01-13 GMT+09:00	1998-07-24	Programmer	IT
4	114	Den Raphaely	1998-03-24 GMT+09:00	1999-12-31	Stock Clerk	Shipping
5	122	Payam Kaufling	1999-01-01 GMT+09:00	1999-12-31	Stock Clerk	Shipping
6	200	Jennifer Whalen	1995-09-17 GMT+09:00	2001-06-17	Administration Assistant	Executive
7	201	Michael Hartstein	1996-02-17 GMT+09:00	1999-12-19	Marketing Representative	Marketing
8	220	Minjun Kim	2015-01-05 GMT+09:00	2019-03-03	Programmer	IT
9	223	Yuna Choi	2021-01-04 GMT+09:00	2023-02-19	Administration Assistant	Administration

[JOIN 10] 데이터 44개

	123 사번(employee_id)	AZ 사원명(employee_name)	AZ 직무명(job_title)	123 급여(salary)	123 직무 최소 급여(min_salary)	123 직무 최대 급여(max_salary)	AZ 급여 적정성
1	100	Steven King	President	24,000	20,000	40,000	범위 내
2	101	Neena Kochhar	Administration Vice President	17,000	15,000	30,000	범위 내
3	102	Lex De Haan	Administration Vice President	17,000	15,000	30,000	범위 내
4	103	Alexander Hunold	Programmer	9,000	4,000	10,000	범위 내
5	104	Bruce Ernst	Programmer	6,000	4,000	10,000	범위 내
6	105	David Austin	Programmer	4,800	4,000	10,000	범위 내
7	106	Valli Pataballa	Programmer	4,800	4,000	10,000	범위 내
8	107	Diana Lorentz	Programmer	4,200	4,000	10,000	범위 내
9	108	Nancy Greenberg	Finance Manager	12,000	8,200	16,000	범위 내
10	109	Daniel Faviet	Accountant	9,000	4,200	9,000	범위 내
11	110	John Chen	Accountant	8,200	4,200	9,000	범위 내
12	111	Ismael Sciarra	Accountant	7,700	4,200	9,000	범위 내
13	112	Jose Manuel Urman	Accountant	7,800	4,200	9,000	범위 내
14	113	Luis Popp	Accountant	6,900	4,200	9,000	범위 내
15	114	Den Raphaely	Purchasing Manager	11,000	8,000	15,000	범위 내
16	115	Alexander Khoo	Purchasing Clerk	3,100	2,500	5,500	범위 내
17	116	Shelli Baida	Purchasing Clerk	2,900	2,500	5,500	범위 내
18	117	Sigal Tobias	Purchasing Clerk	2,800	2,500	5,500	범위 내
19	118	Guy Himuro	Purchasing Clerk	2,600	2,500	5,500	범위 내
20	119	Karen Colmenares	Purchasing Clerk	2,500	2,500	5,500	범위 내
21	120	Matthew Weiss	Stock Manager	8,000	5,500	9,500	범위 내

[VIEW 1] 데이터 45개

	123 사번(employee_id)	AZ 사원명(employee_name)	AZ 이메일(email)	AZ 전화번호(phone_number)	123 입사일(hire_date)	123 급여(salary)
1	100	Steven King	SKING	515.123.4567	2003-06-17	24,000
2	101	Neena Kochhar	NKOCHHAR	515.123.4568	2005-09-21	17,000
3	102	Lex De Haan	LDEHAAN	515.123.4569	2001-01-13	17,000
4	103	Alexander Hunold	AHUNOLD	590.423.4567	2006-01-03	9,000
5	104	Bruce Ernst	BERNST	590.423.4568	2007-05-21	6,000
6	105	David Austin	DAUSTIN	590.423.4569	2005-06-25	4,800
7	106	Valli Pataballa	VPATABAL	590.423.4560	2006-02-05	4,800
8	107	Diana Lorentz	DLORENTZ	590.423.5567	2007-02-07	4,200
9	108	Nancy Greenberg	NGREENBE	515.124.4569	2002-08-17	12,000
10	109	Daniel Faviet	DFAVIET	515.124.4169	2002-08-16	9,000
11	110	John Chen	JCHEN	515.124.4269	2005-09-28	8,200
12	111	Ismael Sciarra	ISCIARRA	515.124.4369	2005-09-30	7,700
13	112	Jose Manuel Urman	JMURMAN	515.124.4469	2006-03-07	7,800
14	113	Luis Popp	LPOPP	515.124.4567	2007-12-07	6,900
15	114	Den Raphaely	DRAPHEAL	515.127.4561	2002-12-07	11,000
16	115	Alexander Khoo	AKHOO	515.127.4562	2003-05-18	3,100
17	116	Shelli Baida	SBAIDA	515.127.4563	2005-12-24	2,900
18	117	Sigal Tobias	STOBIAS	515.127.4564	2005-07-24	2,800
19	118	Guy Himuro	GHIMURO	515.127.4565	2006-11-15	2,600
20	119	Karen Colmenares	KCOLMENA	515.127.4566	2007-08-10	2,500
21	120	Matthew Weiss	MWEISS	650.123.1234	2004-07-18	8,000

[VIEW 2] 데이터 45개

	123 사번(employee_id)	AZ 사원명(employee_name)	AZ 직무명(job_title)	AZ 부서명(department_name)	123 급여(salary)
1	100	Steven King	President	Executive	24,000
2	101	Neena Kochhar	Administration Vice President	Executive	17,000
3	102	Lex De Haan	Administration Vice President	Executive	17,000
4	103	Alexander Hunold	Programmer	IT	9,000
5	104	Bruce Ernst	Programmer	IT	6,000
6	105	David Austin	Programmer	IT	4,800
7	106	Valli Pataballa	Programmer	IT	4,800
8	107	Diana Lorentz	Programmer	IT	4,200
9	108	Nancy Greenberg	Finance Manager	Finance	12,000
10	109	Daniel Faviet	Accountant	Finance	9,000
11	110	John Chen	Accountant	Finance	8,200
12	111	Ismael Sciarra	Accountant	Finance	7,700
13	112	Jose Manuel Urman	Accountant	Finance	7,800
14	113	Luis Popp	Accountant	Finance	6,900
15	114	Den Raphaely	Purchasing Manager	Purchasing	11,000
16	115	Alexander Khoo	Purchasing Clerk	Purchasing	3,100
17	116	Shelli Baida	Purchasing Clerk	Purchasing	2,900
18	117	Sigal Tobias	Purchasing Clerk	Purchasing	2,800
19	118	Guy Himuro	Purchasing Clerk	Purchasing	2,600
20	119	Karen Colmenares	Purchasing Clerk	Purchasing	2,500
21	120	Matthew Weiss	Stock Manager	Shipping	8,000

[VIEW 3]

	AZ 부서명(department_name) ▼	123 부서별 사원 수 ▼	123 부서별 급여 합계 ▼	123 부서별 평균 급여 ▼	123 부서별 최소 급여 ▼	123 부서별 최대 급여 ▼
1	Sales	11	112,000	10,181.82	7,000	14,000
2	Executive	3	58,000	19,333.33	17,000	24,000
3	Finance	6	51,600	8,600	6,900	12,000
4	IT	5	28,800	5,760	4,200	9,000
5	R&D Korea	3	26,800	8,933.33	7,800	10,500
6	Purchasing	6	24,900	4,150	2,500	11,000
7	Shipping	3	24,100	8,033.33	7,900	8,200
8	Accounting	2	20,300	10,150	8,300	12,000
9	Marketing	2	19,000	9,500	6,000	13,000
10	Public Relations	1	10,000	10,000	10,000	10,000
11	Human Resources	1	6,500	6,500	6,500	6,500
12	부서 미배정	1	6,200	6,200	6,200	6,200
13	Administration	1	4,400	4,400	4,400	4,400

[VIEW 4]

	123 관리자 사번(employee_id) ▼	AZ 관리자명(manager_name) ▼	123 관리자별 부하 직원 수 ▼
1	100	Steven King	12
2	145	John Russell	6
3	101	Neena Kochhar	5
4	108	Nancy Greenberg	5
5	114	Den Raphaely	5
6	103	Alexander Hunold	4
7	102	Lex De Haan	2
8	201	Michael Hartstein	2
9	220	Minjun Kim	2
10	205	Shelley Higgins	1

[VIEW 5]

	123 사번(employee_id) ▼	AZ 사원명(employee_name) ▼	🕒 시작일(start_date) ▼	🕒 종료일(end_date) ▼	123 재직 일수(days) ▼	AZ 직무명(job_title) ▼	AZ 부서명(department_name) ▼
1	101	Neena Kochhar	1997-09-21	2001-10-27	1,497	Public Accountant	Accounting
2	101	Neena Kochhar	2001-10-28	2005-03-15	1,234	Accounting Manager	Accounting
3	102	Lex De Haan	1993-01-13	1998-07-24	2,018	Programmer	IT
4	114	Den Raphaely	1998-03-24	1999-12-31	647	Stock Clerk	Shipping
5	122	Payam Kaufling	1999-01-01	1999-12-31	364	Stock Clerk	Shipping
6	200	Jennifer Whalen	1995-09-17	2001-06-17	2,100	Administration Assistant	Executive
7	201	Michael Hartstein	1996-02-17	1999-12-19	1,401	Marketing Representative	Marketing
8	220	Minjun Kim	2015-01-05	2019-03-03	1,518	Programmer	IT
9	223	Yuna Choi	2021-01-04	2023-02-19	776	Administration Assistant	Administration

[MATERIALIZED VIEW 1]

	A-Z 부서명(department_name)	123 부서별 사원 수	123 부서별 급여 합계	123 부서별 평균 급여	123 부서별 최소 급여	123 부서별 최대 급여
1	Sales	11	112,000	10,181.82	7,000	14,000
2	Executive	3	58,000	19,333.33	17,000	24,000
3	Finance	6	51,600	8,600	6,900	12,000
4	IT	5	28,800	5,760	4,200	9,000
5	R&D Korea	3	26,800	8,933.33	7,800	10,500
6	Purchasing	6	24,900	4,150	2,500	11,000
7	Shipping	3	24,100	8,033.33	7,900	8,200
8	Accounting	2	20,300	10,150	8,300	12,000
9	Marketing	2	19,000	9,500	6,000	13,000
10	Public Relations	1	10,000	10,000	10,000	10,000
11	Human Resources	1	6,500	6,500	6,500	6,500
12	부서 미배정	1	6,200	6,200	6,200	6,200
13	Administration	1	4,400	4,400	4,400	4,400

[MATERIALIZED VIEW 2]

	A-Z 국가명(country_name)	123 국가별 사원 수	123 국가별 평균 급여
1	United States	26	8,157.69
2	United Kingdom	12	9,875
3	South Korea	3	8,933.33
4	Canada	2	9,500
5	Germany	1	10,000

[CUBE 1] 데이터 50개

	A-Z 부서명(department_name)	A-Z 직무명(job_title)	123 급여 합계	123 사원 수	A-Z 집계 수준
1	Accounting	Accounting Manager	12,000	1	상세
2	Accounting	Public Accountant	8,300	1	상세
3	Accounting	[전체 직무]	20,300	2	부서별 소계
4	Administration	Administration Assistant	4,400	1	상세
5	Administration	[전체 직무]	4,400	1	부서별 소계
6	Executive	Administration Vice President	34,000	2	상세
7	Executive	President	24,000	1	상세
8	Executive	[전체 직무]	58,000	3	부서별 소계
9	Finance	Accountant	39,600	5	상세
10	Finance	Finance Manager	12,000	1	상세
11	Finance	[전체 직무]	51,600	6	부서별 소계
12	Human Resources	Human Resources Representative	6,500	1	상세
13	Human Resources	[전체 직무]	6,500	1	부서별 소계
14	IT	Programmer	28,800	5	상세
15	IT	[전체 직무]	28,800	5	부서별 소계
16	Marketing	Marketing Manager	13,000	1	상세
17	Marketing	Marketing Representative	6,000	1	상세
18	Marketing	[전체 직무]	19,000	2	부서별 소계
19	Public Relations	Public Relations Representative	10,000	1	상세
20	Public Relations	[전체 직무]	10,000	1	부서별 소계
21	Purchasing	Purchasing Clerk	12,000	5	상세

[CUBE 2] 데이터 31개

	AZ 입사 연도(hire_year)	AZ 커미션 수령 여부	123 인원수	AZ 집계 수준
1	2001	미수령	1	상세
2	2001	[전체]	1	연도별 소계
3	2002	미수령	7	상세
4	2002	[전체]	7	연도별 소계
5	2003	미수령	4	상세
6	2003	[전체]	4	연도별 소계
7	2004	수령	1	상세
8	2004	미수령	2	상세
9	2004	[전체]	3	연도별 소계
10	2005	미수령	8	상세
11	2005	수령	5	상세
12	2005	[전체]	13	연도별 소계
13	2006	수령	2	상세
14	2006	미수령	4	상세
15	2006	[전체]	6	연도별 소계
16	2007	미수령	4	상세
17	2007	수령	2	상세
18	2007	[전체]	6	연도별 소계
19	2008	수령	1	상세
20	2008	[전체]	1	연도별 소계
21	2009	미수령	1	상세

[ROLLUP 1] 데이터 26개

	AZ 국가명(country_name)	AZ 도시(city)	AZ 부서명(department_name)	123 급여 합계	123 사원 수	AZ 집계 수준
1	Canada	Toronto	Marketing	19,000	2	상세
2	Canada	Toronto	[전체 부서]	19,000	2	도시별 소계
3	Canada	[전체 도시]	[전체 부서]	19,000	2	국가별 소계
4	Germany	Munich	Public Relations	10,000	1	상세
5	Germany	Munich	[전체 부서]	10,000	1	도시별 소계
6	Germany	[전체 도시]	[전체 부서]	10,000	1	국가별 소계
7	South Korea	Seoul	R&D Korea	26,800	3	상세
8	South Korea	Seoul	[전체 부서]	26,800	3	도시별 소계
9	South Korea	[전체 도시]	[전체 부서]	26,800	3	국가별 소계
10	United Kingdom	London	Human Resources	6,500	1	상세
11	United Kingdom	London	[전체 부서]	6,500	1	도시별 소계
12	United Kingdom	Oxford	Sales	112,000	11	상세
13	United Kingdom	Oxford	[전체 부서]	112,000	11	도시별 소계
14	United Kingdom	[전체 도시]	[전체 부서]	118,500	12	국가별 소계
15	United States	Seattle	Accounting	20,300	2	상세
16	United States	Seattle	Administration	4,400	1	상세
17	United States	Seattle	Executive	58,000	3	상세
18	United States	Seattle	Finance	51,600	6	상세
19	United States	Seattle	Purchasing	24,900	6	상세
20	United States	Seattle	[전체 부서]	159,200	18	도시별 소계
21	United States	South San Francisco	Shipping	24,100	2	상세

[ROLLUP 2] 데이터 46개

	○ AZ 입사 연도(hire_year) ▼	AZ 부서명(department_name) ▼	123 인원수 ▼	123 급여 합계 ▼	AZ 집계 수준 ▼
1	2001	Executive	1	17,000	상세
2	2001	[전체 부서]	1	17,000	연도별 소계
3	2002	Accounting	2	20,300	상세
4	2002	Finance	2	21,000	상세
5	2002	Human Resources	1	6,500	상세
6	2002	Public Relations	1	10,000	상세
7	2002	Purchasing	1	11,000	상세
8	2002	[전체 부서]	7	68,800	연도별 소계
9	2003	Administration	1	4,400	상세
10	2003	Executive	1	24,000	상세
11	2003	Purchasing	1	3,100	상세
12	2003	Shipping	1	7,900	상세
13	2003	[전체 부서]	4	39,400	연도별 소계
14	2004	Marketing	1	13,000	상세
15	2004	Sales	1	14,000	상세
16	2004	Shipping	1	8,000	상세
17	2004	[전체 부서]	3	35,000	연도별 소계
18	2005	Executive	1	17,000	상세
19	2005	Finance	2	15,900	상세
20	2005	IT	1	4,800	상세
21	2005	Marketing	1	6,000	상세